

SOFTWARE REQUIREMENTS SPECIFICATION

AI-BASED FAKE PROFILE DETECTION IN SOCIAL NETWORKS

1. INTRODUCTION

This Software Requirements Specification defines the functional, non-functional, interface, data, security and operational requirements of the proposed fake-profile detection system.

2. SYSTEM PURPOSE

The system analyzes structured account information and predicts whether a profile is likely to be genuine or suspicious. It generates a probability, risk category, and explanation.

3. FUNCTIONAL REQUIREMENTS

FR-01: User authentication.

FR-02: Profile data submission.

FR-03: Input validation.

FR-04: Missing-value and format handling.

FR-05: Behavioral feature generation.

FR-06: Fake/genuine prediction.

FR-07: Risk score generation.

FR-08: Explanation generation.

FR-09: Prediction history.

FR-10: Analyst review.

FR-11: Report export.

FR-12: Model-version tracking.

FR-13: Administrative monitoring.

4. NON-FUNCTIONAL REQUIREMENTS

Performance: The prediction API should provide low-latency responses under expected load.

Reliability: Service failures should return controlled errors.

Security: Authentication, authorization, input validation and secure secret storage are required.

Scalability: API and model-serving components should be independently scalable.

Maintainability: Components should have clear responsibilities.

Usability: Risk results should be understandable to non-ML analysts.

Explainability: Important contributing features should be visible.

5. HARDWARE REQUIREMENTS

Development machine: 8 GB RAM or more, modern CPU, 10 GB free storage. GPU is optional.

6. SOFTWARE REQUIREMENTS

Python 3.x, Pandas, NumPy, Scikit-learn, FastAPI, Uvicorn, React, database server, Git and Docker.

7. USER ROLES

Administrator: manages configuration and model versions.

Analyst: analyzes and reviews suspicious accounts.

System: executes preprocessing, prediction and scoring.

8. DATA REQUIREMENTS

Profile information, activity statistics, labels for training, prediction records, review records and model metadata.

9. SECURITY REQUIREMENTS

Sensitive identifiers should be minimized or hashed. API requests must be validated. Administrative operations require authorization. Logs should avoid unnecessary personal information.

10. ACCEPTANCE CRITERIA

The system is acceptable when valid profile data produce a prediction, invalid data are rejected safely, the dashboard displays the result correctly, and model performance is measured on unseen test data.